

Virtual Reality



Image sources: <https://www.iotforall.com/current-state-augmented-virtual-reality/>, <https://www.aniwaa.com/best-of/vr-ar/best-smartphone-vr-headset/>, <https://www.pcgamer.com/best-vr-headset/>, <https://media.direct.playstation.com/is/image/psdglobal/PSVR2-Hero>

The concept of presence



- The feeling you are actually some place else
 - Reaching out for a virtual object
 - Dodging virtual obstacles
 - Disoriented when returning to the real world
- For many people, presence is simply magic!



Image sources: https://www.overclock3d.net/news/gpu_displays/amd_announces_new_liquidvr_technology/1, <https://www.nature.com/articles/nrn1651>

Simulation Sickness



- Also known as **motion sickness**
 - Disagreement exists between visually perceived movement and the vestibular system's sense of movement.



www.space-junk-comics.com



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Image sources: <https://www.familiesgotravel.com/2017/01/8-tips-car-sick-kids/>, https://66.media.tumblr.com/5368851a1d03aaf8cf70801f02514643/tumblr_orhqalv2lF1uel1m3o1_1280.jpg

Simulation Sickness Theories



- **Cue Conflict Theory**
 - Mismatch between what we expect to see and feel and what we are actually seeing and feeling.
- **Postural Instability Theory**
 - Our body is constantly making small adjustments to not fall over and relies on accurate sensory input to do so.
- **Poison Theory**
 - Evolutionary: One of the only times you will experience sensory input that is different than what you expect, or when you get all dizzy and have problems standing up, is when you have been poisoned.

Simulation Sickness - Development Considerations



- Potential causes of simulation sickness in VR
 - **A: Player is sitting still, while the avatar is moving**
 - Acceleration, deceleration, rotational movements, etc.
 - **B: Showing the same static image over multiple frames**
 - Head movements have no effect
 - **C: Correct tracking with too much latency**
 - Brain does not accept this as natural
- Developers have (limited) control over **A** and **B**, while **C** is mostly caused by hardware limitations!

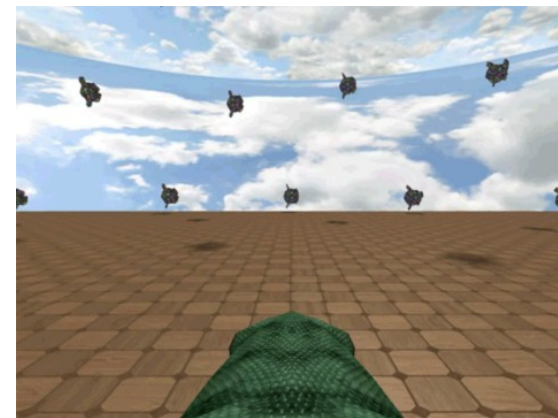


Image sources: http://www.gamasutra.com/blogs/BenLewisEvans/20140404/214732/Simulation_Sickness_and_VR_What_is_it_and_what_can_developers_and_players_do_to_reduce_it.php

Frame Rate and Latency (1/2)



- Frame Rate describes the number of frames rendered (per second)
 - VSync is required in VR (typically 75, 90, 120 Hz)
 - Screen tearing becomes “world tearing” in VR!
 - FPS alone is not enough information to determine VR latency
 - Command buffering, double/triple buffering, post image processing, etc.
- Latency (overall latency)
 - Time from physical input to photon on retina → Motion-to-Photon (M2P)
 - Measured in milliseconds, maximal value is relevant
 - Displays “scan” the front buffer line-wise
 - app-to-photon-start versus app-to-photon-end → app-to-mid can be used
 - VR headsets typically have a photodiode to measure latency

=> Goal [for latency] is < 20 ms, and hopefully close to 5 ms

Frame Rate and Latency (2/2)



- Causes of latency
 - Input sensor processing
 - Simulation of the game loop
 - Frame-queueing (command buffer on GPU)
 - Rendering (processing of draw calls)
 - VSync (waiting for next frame)
 - Image transmission (ports have limited bandwidth)
 - Display electronics (post-processing and scanning)
 - Pixel-switching
 - TFT is much slower than OLED

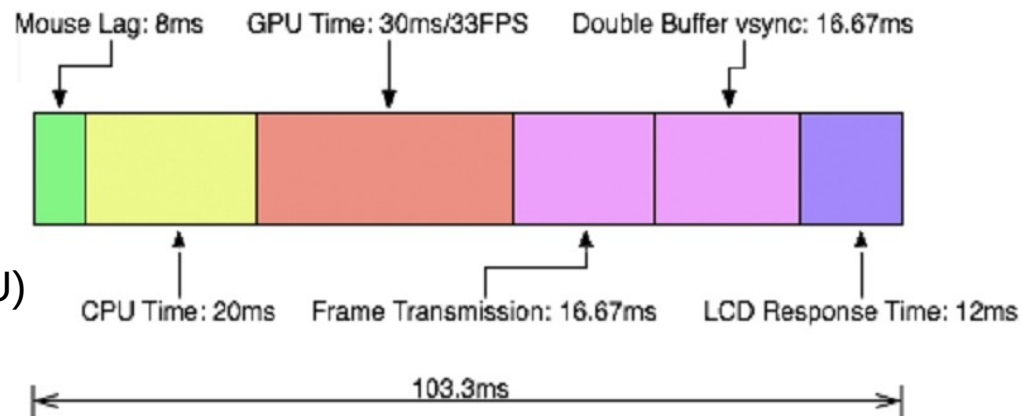


Image sources: Untersuchung aktueller Virtual Reality Technologien am Beispiel einer Applikation zur Visualisierung und Konfiguration eines Fahrzeugs (Bachelor Thesis), Rupert Jung, Hochschule der Medien, 2015

Hardware Setup (1/4)



- Displays and resolution
 - OLED (in contrast to TFT) has higher contrast (black) and is faster
 - Human eye has a angular resolution of 0.07°
 - Theoretically, 576 Mega pixel @ FOV 120° FOV are required
 - Ironically, low resolution has little impact on the experienced presence
 - The pixel density of the display needs to be high (pixel screendoor)

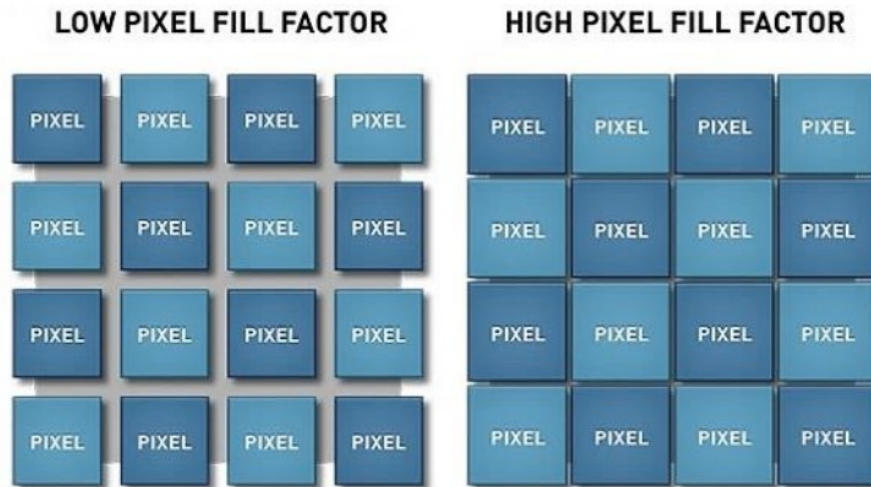


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Hardware Setup (2/4)



- Typical setup of HMDs
- Problems
 - Geometric distortion
 - Chromatic aberration
 - Focus point is only in the middle

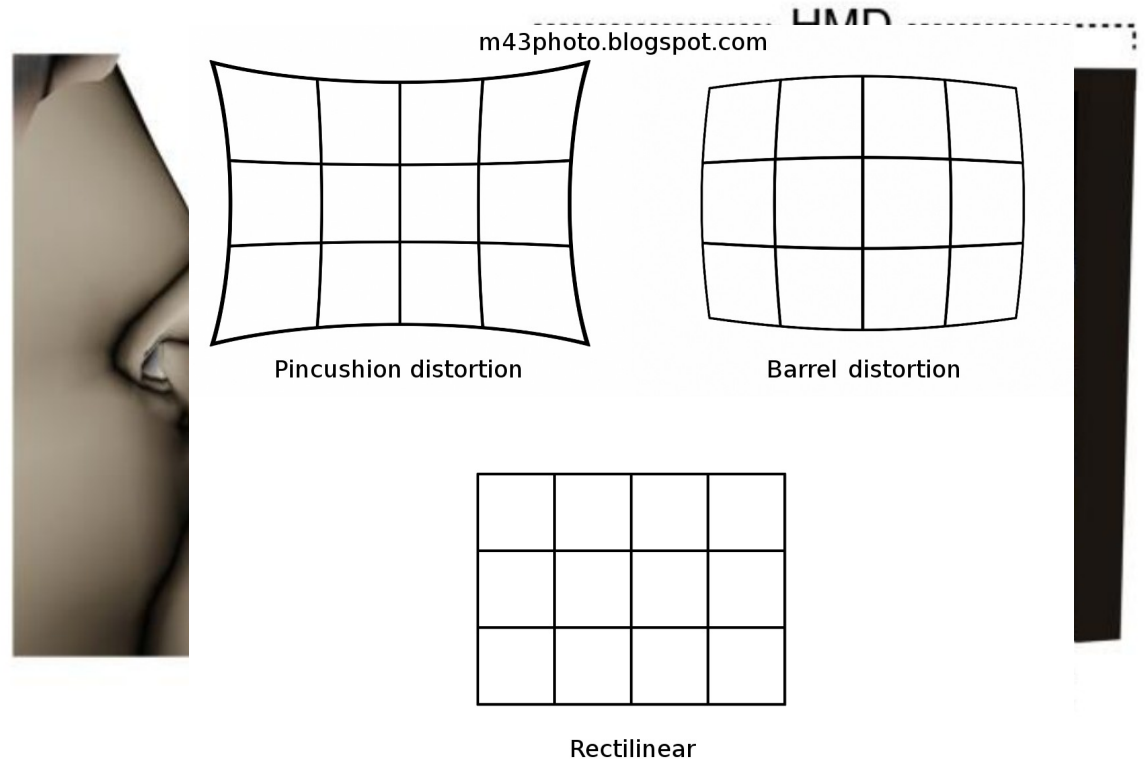


Image sources: Untersuchung aktueller Virtual Reality Technologien am Beispiel einer Applikation zur Visualisierung und Konfiguration eines Fahrzeugs (Bachelor Thesis), Rupert Jung, Hochschule der Medien, 2015
<https://s.studiobinder.com/wp-content/uploads/2022/08/What-is-Chromatic-Aberration-Chromatic-aberration-explained.png>

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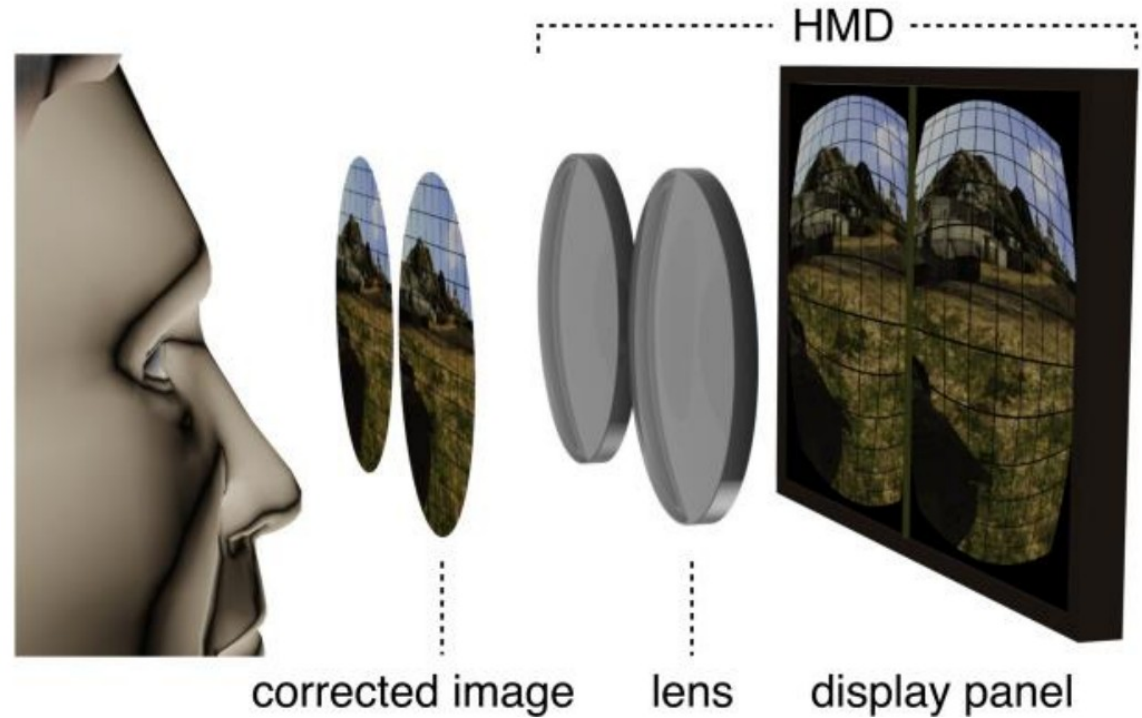


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Hardware Setup (2/4)



- Typical setup of HMDs

- Problem

- Geometric
- Chromatic
- Focus

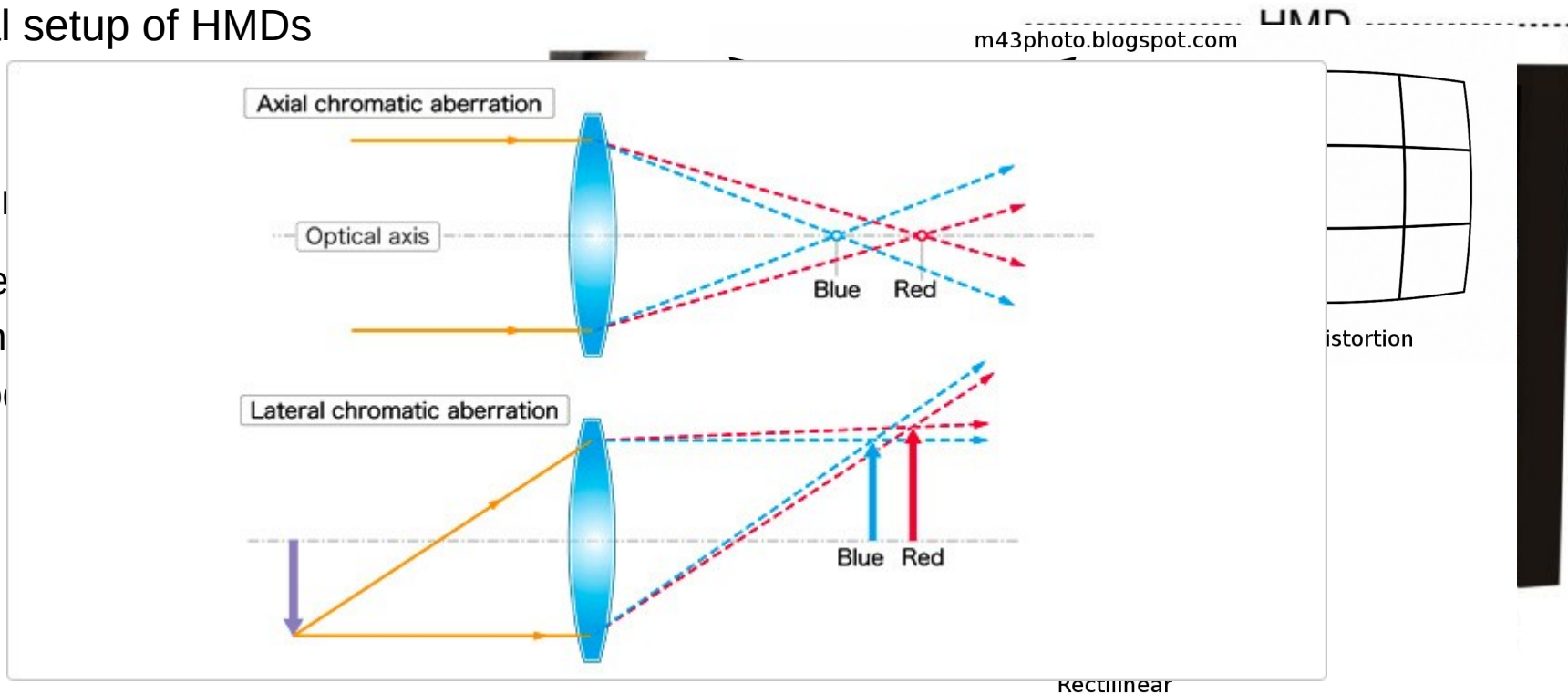


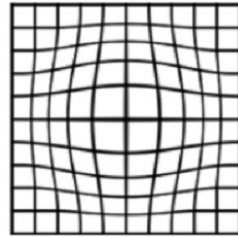
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Hardware Setup (3/4)

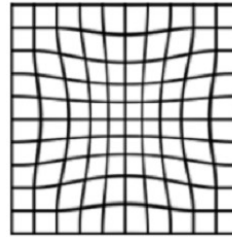


- Barrel Distortion

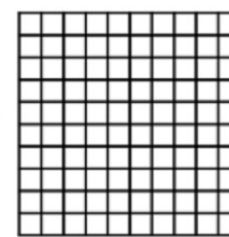
Barrel Distortion
(In-Engine)



Pin-cushion Distortion
(From Rift Lenses)



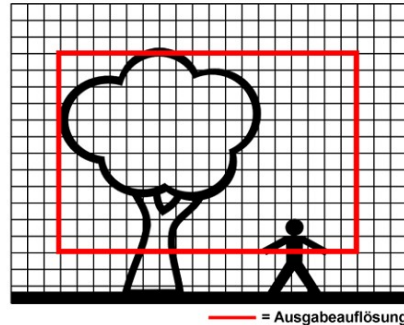
No Distortion
(Final Observed Image)



=> Higher rendering resolution is required! (~1.5-2.0 times)

- Image calibration (to correct chromatic aberration)

Renderbuffer (intern)



Bildaufbereitung (Frontbuffer)

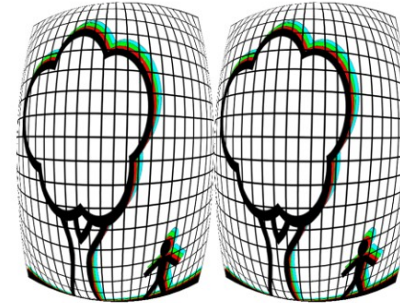


Image sources: Untersuchung aktueller Virtual Reality Technologien am Beispiel einer Applikation zur Visualisierung und Konfiguration eines Fahrzeugs (Bachelor Thesis), Rupert Jung, Hochschule der Medien, 2015

Hardware Setup (4/4)



- **Head tracking**

- Limited by latency, noise, and drift
- **Rotational tracking** (3 degrees of freedom)
 - Accelerometer, gyroscope, magnetometer
- **Positional tracking** (3 degrees of freedom)
 - Inside-out: HMD tracks environment
 - Outside-in: External sensor tracks HMD

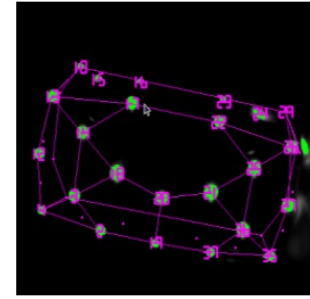
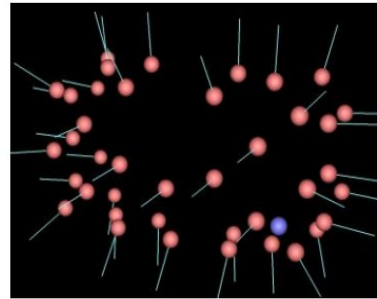
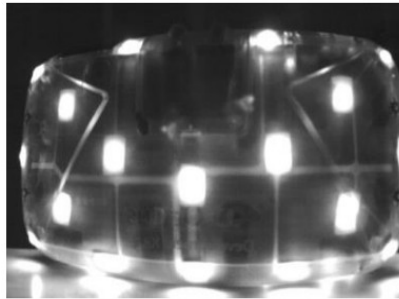
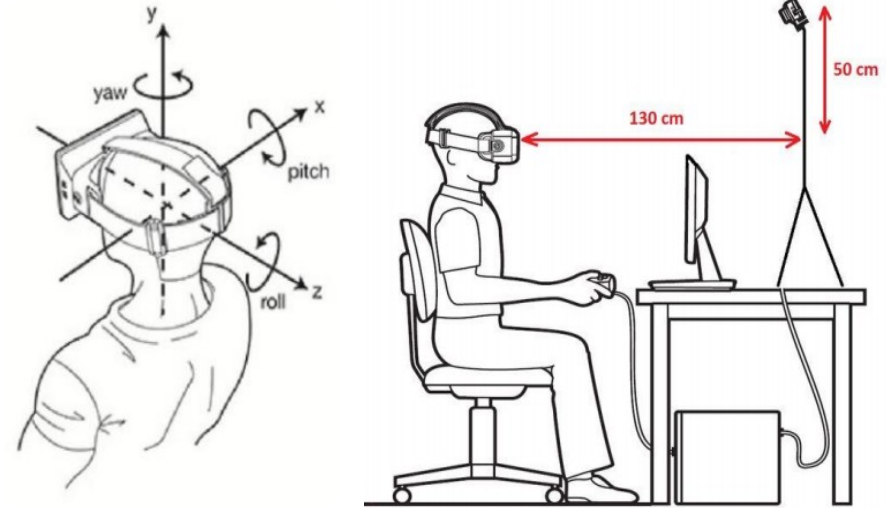


Image sources: Untersuchung aktueller Virtual Reality Technologien am Beispiel einer Applikation zur Visualisierung und Konfiguration eines Fahrzeugs (Bachelor Thesis), Rupert Jung, Hochschule der Medien, 2015

Head Tracking (1/3)



- Positional tracking using external sensors or trackers

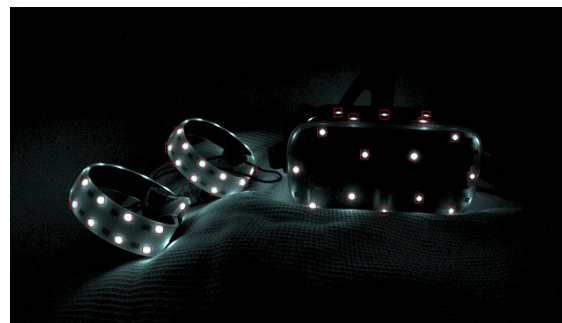
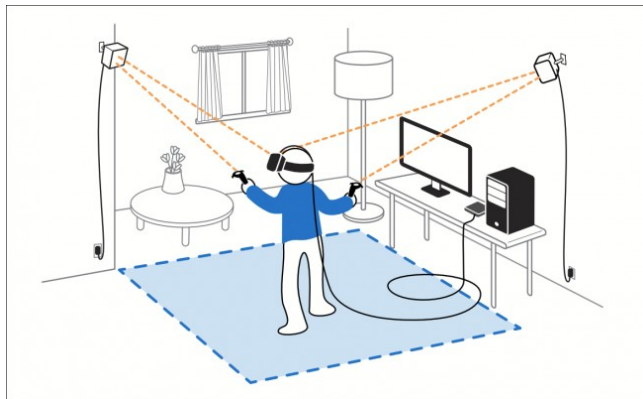


Image sources: <https://www.howtogeek.com/246333/oculus-rift-vs.-htc-vive-which-vr-headset-is-right-for-you/>, <https://www.vrheads.com/least-painful-way-set-htc-vive-lighthouses>, <https://www.roadtovr.com/oculus-ac-knowledges-ongoing-tracking-issues-3-sensor-setups-promises-february-update/>, <http://www.openhmd.net/index.php/category/oculus-rift/>

Head Tracking (2/3)



- Positional tracking: HTC Vive Pro / Valve Index
 - Backwards compatible with HTC Vive trackers and controllers
 - Full support for second-generation Lighthouse 2.0 trackers
 - Up to four base stations for more accurate tracking
 - Tracking area up to 100 meters squared (33 x 33 feet)
 - (also wireless, based on Intel's low-latency WiGig technology)



Image sources: <https://www.wired.com/review/review-htc-vive-pro/>

Head Tracking (3/3)



- Positional tracking: (old) Oculus Rift / PlayStation VR

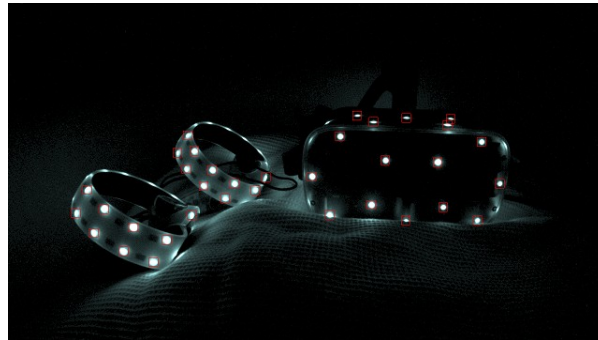


Image sources: <https://variety.com/2018/digital/news/playstation-vr-headset-price-cut-1202702214/>, <https://www.wareable.com/vr/field-of-view-sony-playstation-vr-move-controller-patent-123>

Positional tracking using internal cameras



- Oculus/Meta Rift/Quest, Windows Mixed Reality, HoloLens, Playstation VR 2, (Apple Vision Pro), ...
- “True” **inside-out** tracking with no base stations
 - Multiple (visible-light, low-resolution, black and white) cameras to observe features in the environment
 - Fused with Inertial Measurement Unit (IMU) data to determine a precise position of the device in the environment
- Downsides
 - Sufficient environmental features are required
 - Also, the environment might change unexpectedly...
 - Enough light is needed (may not work in the dark)
 - Controllers need to be in the cameras' field of view

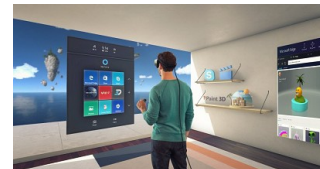


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Eye Tracking in VR



- How does it work?
 - Measuring the distance between the pupil center and the reflection of the cornea
 - Infrared light, invisible to the human eye, creates this reflection while cameras record and track the movements
 - Special challenge in VR:
Vergence distance $\neq$ focal distance
- What are the benefits?
 - Foveated rendering: Reducing the image quality in the peripheral vision
 - Enhanced visual realism
 - Better depth perception
 - Gaze-based interactions

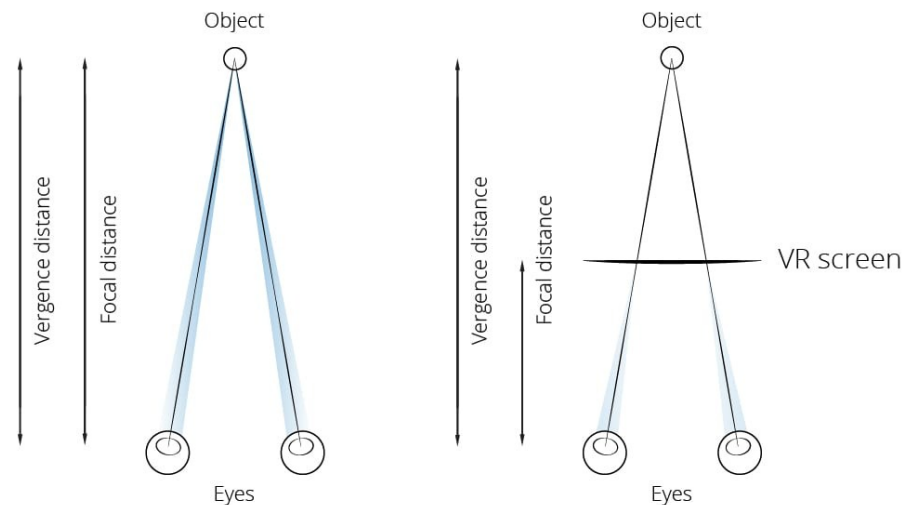
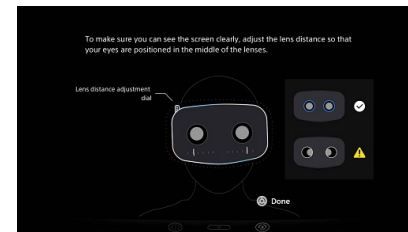
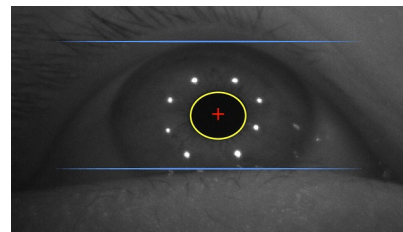


Image sources: <https://www.roadtovr.com/wp-content/uploads/2017/04/eye-tracking-aglass.jpg>, <https://always-on.com.au/wp-content/uploads/2023/02/PSVR2-Eye-Tracking-2-scaled.jpg>, <https://imotions.com/blog/learning/best-practice/vr-eye-tracking/>

Optimization Techniques (1/7)



- Low Persistence
 - Eyes automatically correct the head movement (vestibulo-ocular reflex)
=> With an HMD, the image “smears” over the retina (strobing)
 - A low persistence display flashes quickly, then shows a black image

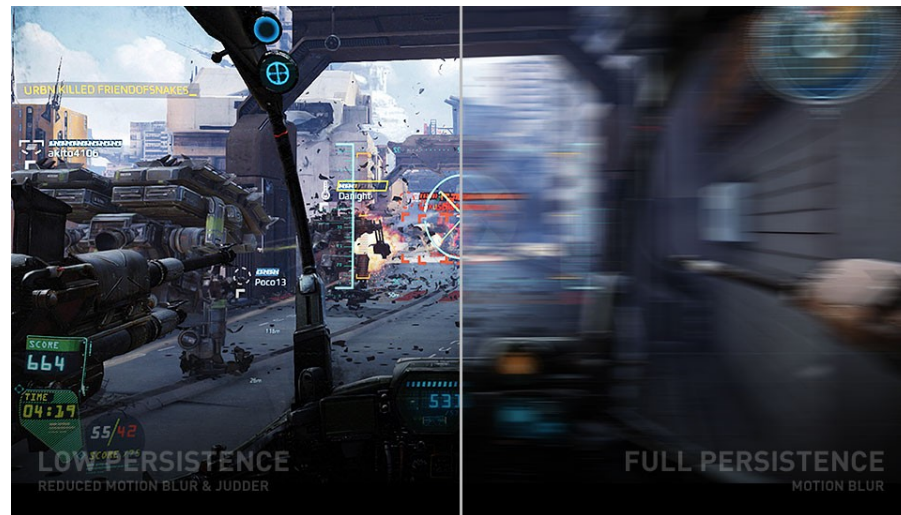
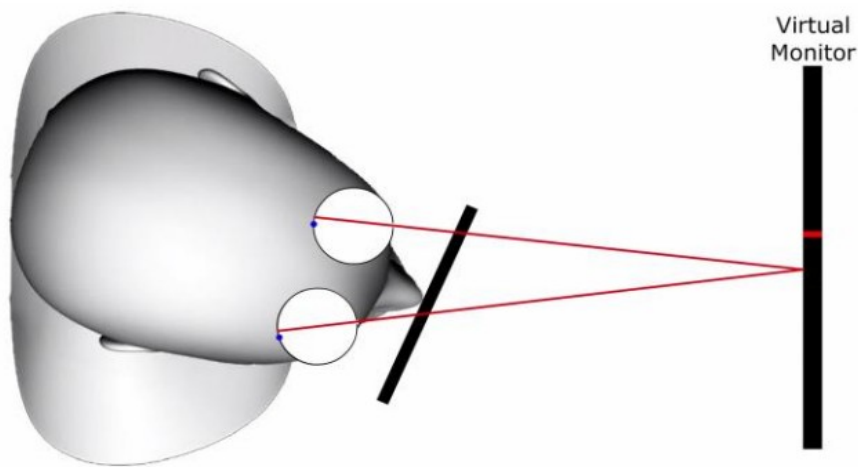


Image sources: Untersuchung aktueller Virtual Reality Technologien am Beispiel einer Applikation zur Visualisierung und Konfiguration eines Fahrzeugs (Bachelor Thesis), Rupert Jung, Hochschule der Medien, 2015, <https://www.extremetech.com/gaming/178839-oculus-rift-dk2-goes-on-sale-for-350-features-low-latency-1080p-displays-more-polished-appearance>

Optimization Techniques (2/7)



- Low persistence displays

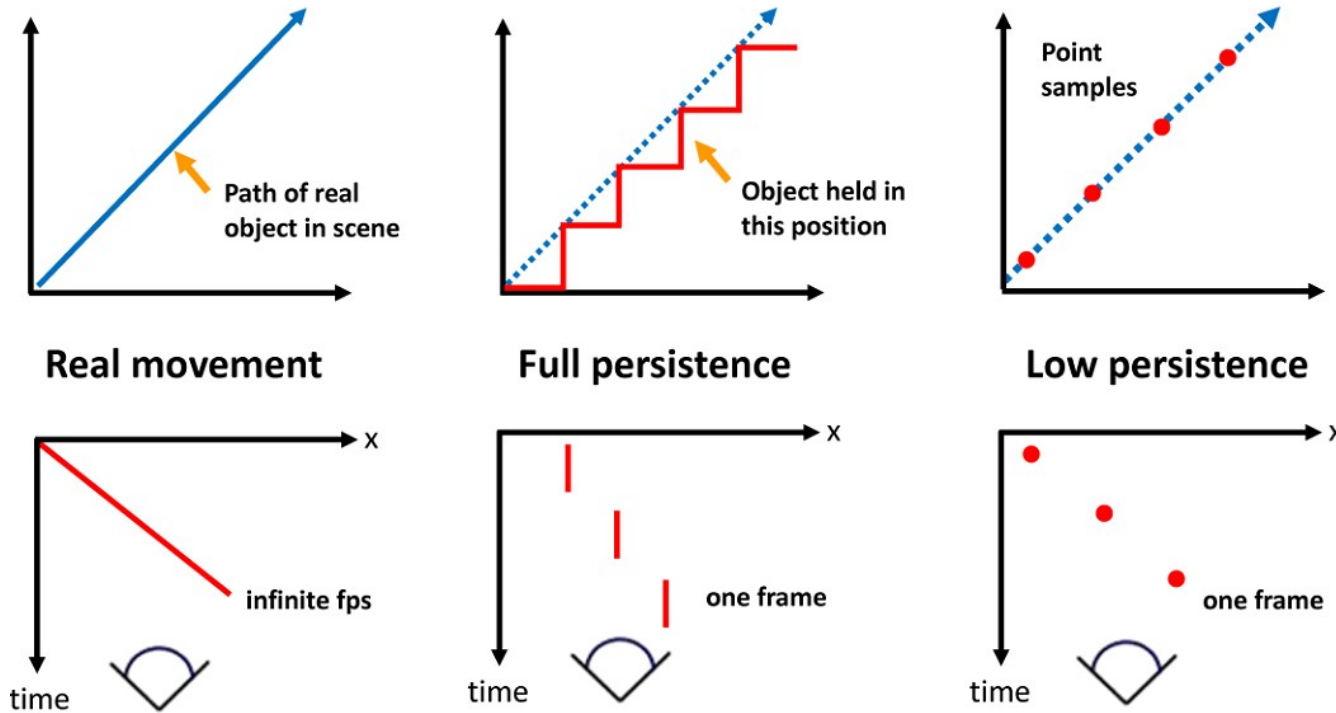


Image sources: Untersuchung aktueller Virtual Reality Technologien am Beispiel einer Applikation zur Visualisierung und Konfiguration eines Fahrzeugs (Bachelor Thesis), Rupert Jung, Hochschule der Medien, 2015

Optimization Techniques (3/7)



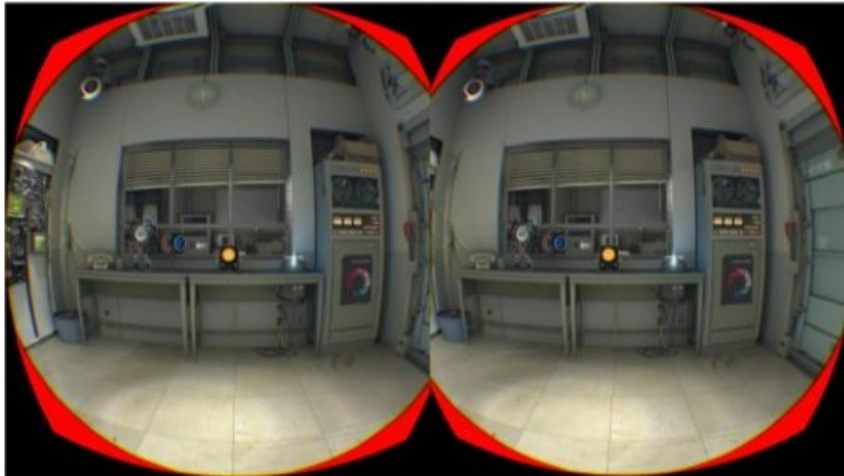
- Dynamic Motion Prediction (DMP)
 - Current image is created based on “old” data
 - Mostly importantly the camera matrices
 - Idea: Use the natural inertia of the human head to form predictions
 - The head will not instantly invert its motion
 - Accuracy of DMP depends on...
 - Quality and noise of the sensor data
 - Prediction strategy (angular acceleration versus angular velocity)
 - Underlying filtering model (e.g. Bayes, Kalman)
 - Number and frequency of measurements
 - Length of the prediction interval
 - Inertial Measurement Unit (IMU) operates with ~1kHz
 - A large number of measurements is available
 - Prediction intervals can be reasonably short

Optimization Techniques (4/7)



- Stencil Rejection / Hidden Area Mesh
 - Stencil out pixels which cannot be seen anyway
 - Roughly 17% fill rate can be saved

Wasted pixels



Hidden area mesh



Image sources: Untersuchung aktueller Virtual Reality Technologien am Beispiel einer Applikation zur Visualisierung und Konfiguration eines Fahrzeugs (Bachelor Thesis), Rupert Jung, Hochschule der Medien, 2015

Optimization Techniques (5/7)



- Multi-Resolution Shading / Multi-Projection

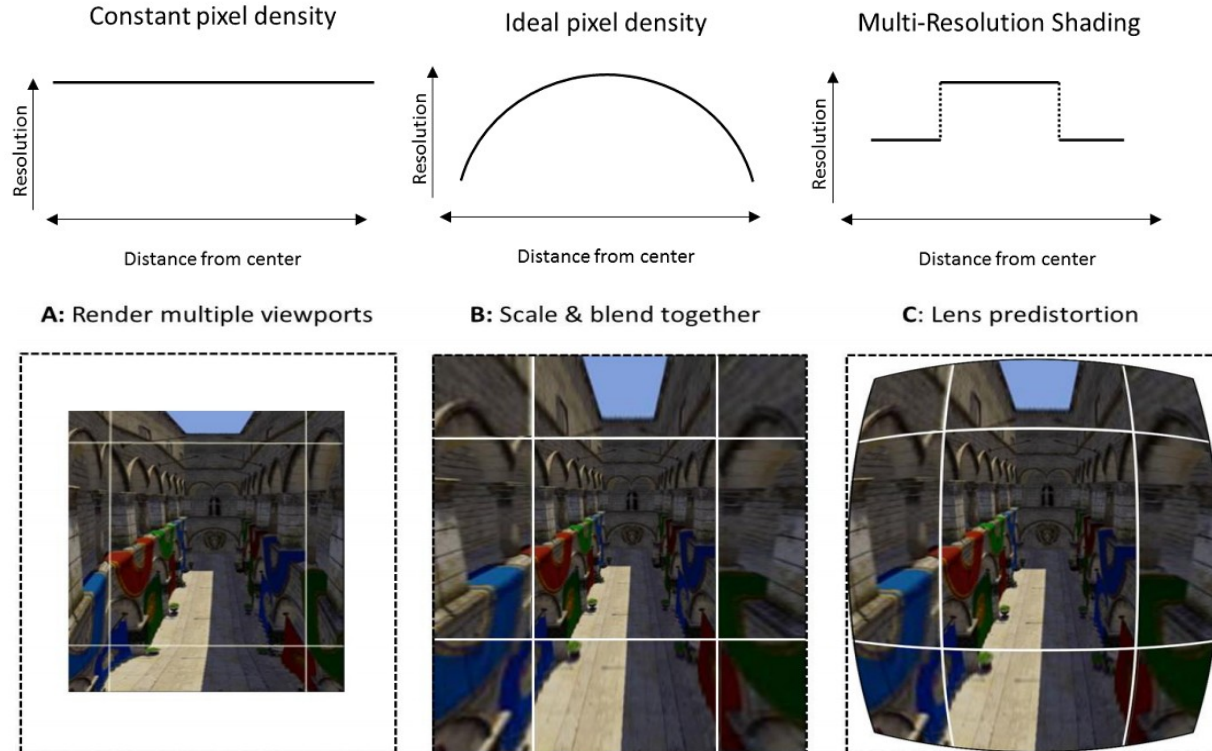


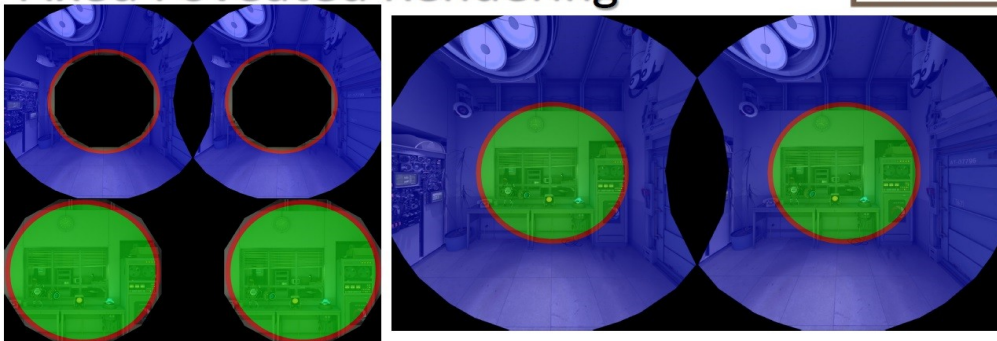
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Optimization Techniques (6/7)



Fixed Foveated Rendering

VALVE



Adaptive Quality

Quality Level	MSAA	Resolution Scale	Render Resolution
+6	8x	1.4	2116x2352
+5	8x	1.3	1965x2184
+4	8x	1.2	1814x2016
+3	8x	1.1	1663x1848
+2	8x	1.0	1512x1680
+1	4x	1.1	1663x1848
0	4x	1.0	1512x1680
-1	4x	0.9	1360x1512
-2	4x	0.81	1224x1360
-3	4x	0.73	1102x1224
-4	4x	0.65	992x1102

Radial Density Masking

VALVE

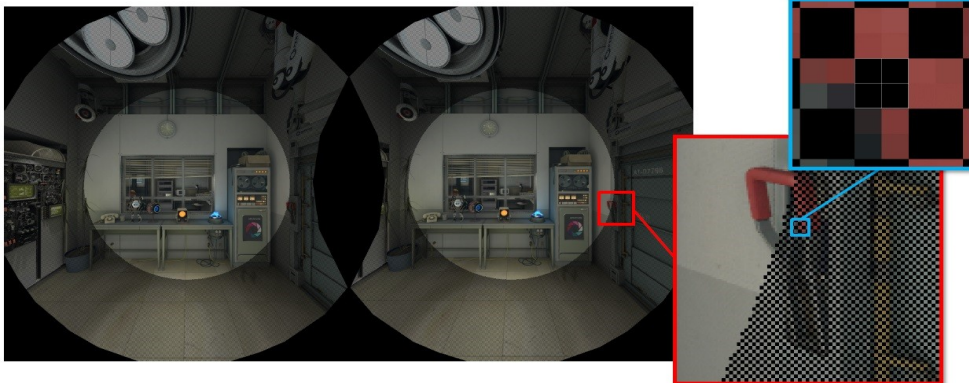


Image sources: <https://arstechnica.com/gaming/2016/03/how-valve-got-passable-vr-running-on-a-four-year-old-graphics-card/>

Optimization Techniques (7/7)



- Queue-Ahead / Running Start
 - To guarantee low latency, CPU and GPU must be synchronized
 - Problem: Issuing draw calls takes time and wastes resources

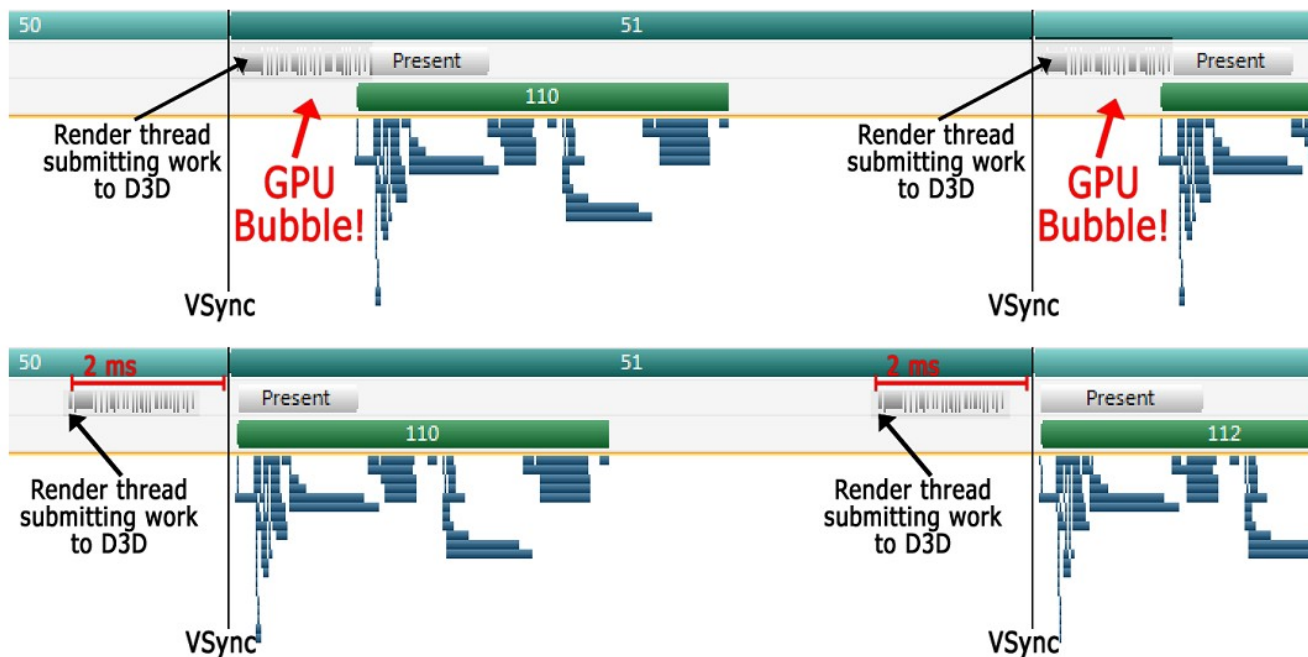
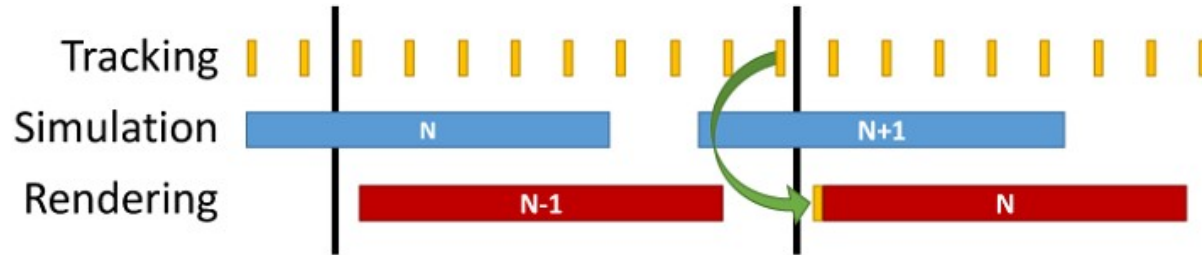


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Advanced Optimization Techniques (1/3)



- Late-Latching
 - The camera view matrix is not calculated when sending the commands to the GPU, but when the first command is processed.
 - The headset constantly updates the matrix based on the newest sensor data and the GPU can read the data directly from the RAM.
 - Similar principle is used for a hardware mouse cursor.



- Caveats and limitations
 - Commands are still buffered, only the view matrix is “up-to-date”
 - The CPU simulation cannot assume a particular view matrix
 - Shader code must be rewritten to use the proper view matrix
 - Culling mechanisms must be more “tolerant”

Image sources: Untersuchung aktueller Virtual Reality Technologien am Beispiel einer Applikation zur Visualisierung und Konfiguration eines Fahrzeugs (Bachelor Thesis), Rupert Jung, Hochschule der Medien, 2015

Advanced Optimization Techniques (2/3)



- Timewarp
 - Correct head rotation is crucial and late-latching alone is not enough to satisfy the latency requirement of < 20 ms
 - Idea: Modify the scene after rendering
 - Pixels are shifted in a post processing step
 - Problems
 - Occlusion information is not available
 - (Thus) only applicable to rotation
 - Black borders in the peripheral vision
 - => The frame rate must be high enough!
 - Early and late display pixels have different latencies

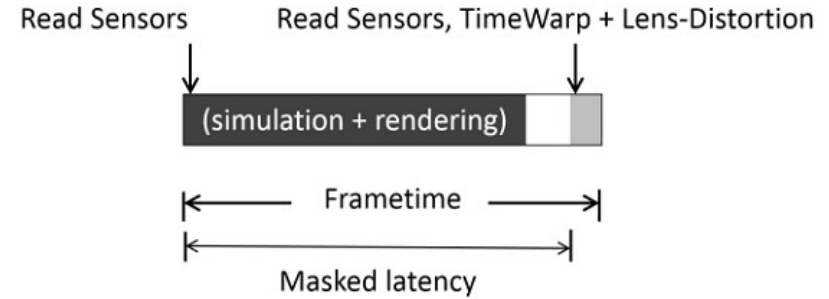
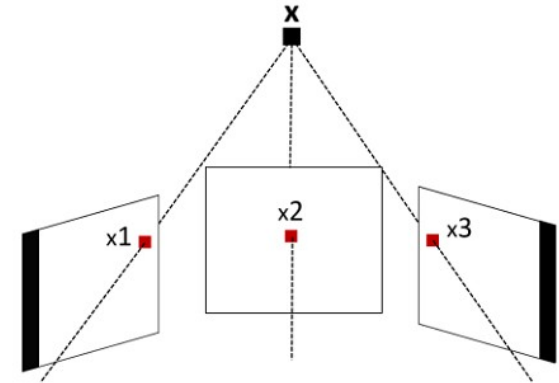
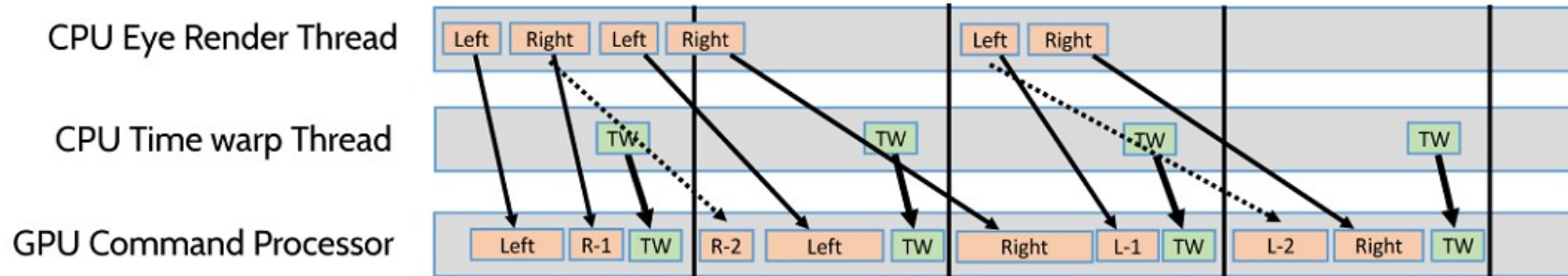


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Advanced Optimization Techniques (3/3)



- Asynchronous Timewarp
 - Like Timewarp, but on another thread in parallel (i.e. asynchronously) with rendering.
 - Framerate and display frequency can be decoupled!
 - Thus render times are more flexible and the hardware is utilized better.



- GPU Priorities / Preemption
 - Application must be able to pause the rendering, to update a new frame at any time.
 - May not be available on every platform and for every graphics driver.

Finally! VR Works!



- We have come a long way...



Image sources: <https://blogs.haverford.edu/decentered/2015/08/04/mass-media-and-the-rhetoric-of-technological-progress/>, <http://www.nebeep.com/gamerama-and-occulus/>

Valve: Advanced VR Rendering Performance



- But wait... a video from 2016!?
 - All of this information is still relevant and up-to-date
 - The only real advancement worth adding is Foveated Rendering (FR)
 - FR combines eye tracking with multi resolution shading
 - However, hardware availability and added latency are persistent issues



https://www.youtube.com/watch?v=DdL3WC_oBO
4

References



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